MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER

TYPE 7843

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the electron tube described herein shall consist of this specification and the latest issue of MIL-E-1.

DESCRIPTION: Tetrode, UHF, ceramic-metal
See figure 1
Mounting position: Any
Weight: 2 ounces nominal (56.7 grams)

ABSOLUTE RATINGS: F = 400 MHz

Parameter:	Ef	Eb	Ec1	Ec2	Ib	Ic1	Pg1	Pg2	Pp
Unit:	V (Note 1)	V dc	V dc	V dc	mA dc	mA dc	W	W	W
Maximum:									
AB audio:	26.5 ±10%	1,000	---	300	180	30	1.0	7.0	115
AB SSBSC:	26.5 ±10%	1,000	---	300	250 (Note 2)	30	1.0	7.0	115
C telep:	26.5 ±10%	800	-100	300	150	30	1.0	4.6	75
C teleg:	26.5 ±10%	1,000	-100	300	180	30	1.0	7.0	115
TEST CONDITIONS:	26.5	1,000	Adj	300	---	---	---	---	---

ABSOLUTE RATINGS: F = 400 MHz

Parameter:	Pi	T(seal)	tk	Cooling	Barometric pressure, reduced
Unit:	W	°C (Note 3)	sec (Min)	--- (Note 4)	mmHg
Maximum:					
AB audio:	180	250	60	---	141.2
AB SSBSC:	180	250	60	---	141.2
C telep:	120	250	60	---	141.2
C teleg:	180	250	60	---	141.2
TEST CONDITIONS:	---	---	60	Note 5	---

GENERAL:

Qualification - Required

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Method	Requirement or test	Notes	Conditions	Symbol	Limits Min	Limits Max	Unit
	Quality conformance inspection, part 1	6					
1266	Total grid current	7,8	Ec1/Ib = 115 mA dc	Ic1	---	-8.0	µA dc
1256	Electrode current (screen)	-	Ec1/Ib = 115 mA dc	Ic2	-4.7	2.0	mA dc
1261	Electrode voltage (1) (grid)	-	Ec1/Ib = 115 mA dc	Ec1	-9.0	-18.0	V dc
1261	Electrode voltage (2) (grid)	-	Ec1/Ib = 1.0 mA dc	Eco	---	-48	V dc
1231	Pulsing emission	9	etd/ik = 10 a	etd	---	300	v
---	Zero-bias anode current	10	Ef = 21.5 V; Eb = 400 V dc; Ec2 = 200 V dc; Ec1 = 0; tk = 120; t = 60 (max)	Ibo	170	---	mA dc
1301	Heater current	-		If	0.48	0.60	A
	Quality conformance inspection, part 2						
1266	Primary grid emission (control)	-	Eg1/g1 input = 2 W; t = 30; anode and g2 grounded	Isg1	---	-2.0	µA dc
1266	Primary grid emission (screen)	-	Eg2/g2 input = 7.0 W; t = 30; anode and g1 grounded	Isg2	---	-3.0	µA dc
1366	Interelement leakage resistance, cold	11	Supply voltage = 200 V dc	R	1.0	---	Meg
2214	Power output	12	F = 400 ±20 MHz; Ib = 180 mA dc (max); Ic1 = 30 mA dc (max); t = 120; Pd = 3.3 W (max); Ef = 24.0 V	Po	85	---	W
1331	Direct-interelectrode capacitance	-	Use capacitance fixture per 289-JAN	Cg1k	11.0	15.0	pF
				Cg1g2	15.0	20.0	pF
				Cg2p	4.2	5.2	pF
				Cg1p	---	0.065	pF
				Cpk	---	0.013	pF
				Cg2k	0.20	0.45	pF
1236	Power oscillation	-	F = 15 MHz; Ef = 21.5 V; Eb = 850 V dc; Ec2 = 300 V dc (max); Ib = 150 mA dc; Rg1 = 4,000 ohms (max); Ic1 = 30 mA dc; t = 120	Po	80	---	W (use ful)
1372	Current division, long pulse (method A)	-	Eb = 350 V dc; Ec1 = -100 V dc; egk/ib = 0.6 a; prr = 10 to 12; tp = 4,500 to 5,000 µs	egk	-3.0	+17	v
				ic1	0	130	ma
				ic2	10	70	ma

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	Method	Requirement or test	Notes	Conditions	Symbol	Limits Min	Limits Max	Unit
		Quality conformance inspection, part 3						
Ⓓ	---	Life test	8,12	Group C; Ef = 26.0 V; Ib = 180 mA dc; F = 400 ±20 MHz; Ic1 = 30 mA dc (max); Ec1 = value obtained for Eco in electrode voltage (2) grid test per part 1 herein; t = 500 hours	---	---	---	---
	---	Life-test end points:						
	1231	Pulsing emission	-	etd/ik = 7.5 a	etd	---	400	v
	1266	Primary grid emission (control)	-		Isg1	---	-4.0	µA dc
	1266	Primary grid emission (screen)	-		Isg2	---	-4.0	µA dc
	2214	Power output	-		Po	75	---	W
	1236	Power oscillation	-		Po	65	---	W (use-ful)
	1506	Heater-cycling life	-	Group C; Ef = 35.0 V; accelerated heater-cycling 2-1/2 minutes "on" 5 minutes "off"	---	384	---	Cycles
	---	Heater-cycling life-test end point:						
	1301	Heater current	-		If	0.4	0.7	A
Ⓓ	1031	Low-frequency vibration	13	Ebb = 300 V dc; Ec2 = 250 V dc; Ec1/Ib = 10 mA dc; Rp = 2,000 ohms	Ep	---	300	mV ac
Ⓓ	1031	High-frequency vibration	13	No voltages	---	---	---	---
Ⓓ	1042	Shock, specified pulse	13	15 G, 11 ms, half-sine; no voltages	---	---	---	---
Ⓓ	---	Vibration and shock-test end points:						
	1266	Total grid current	8	Ec1/Ib = 115 mA dc	Ic1	---	-8.0	µA dc
	1261	Electrode voltage (1) (grid)	-	Ec1/Ib = 115 mA dc	Ec1	-9.0	-18.0	V dc
	1002	Barometric pressure, reduced	13	Pressure = 141.2 mmHg; voltage = 1,300 V dc applied between p and k, with k, g1, and g2 tied together	---	---	---	---

NOTES:

1. Because the cathode is subjected to considerable back bombardment as the frequency is increased, with resultant increase in temperature, the heater voltage shall be decreased depending on operating conditions and frequency to prevent overheating the cathode and resultant short life.

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NOTES: - Continued

2. The maximum dc anode current at the peak of the current curve is 250 mA dc for a signal having a minimum peak-to-average power ratio of 2.0. The maximum rating for a signal having a minimum peak-to-average power ratio of less than 2.0, such as is obtained in single-tone operation, is 180 mA dc. During short periods of circuit adjustment under single-tone conditions, the average anode current may also reach the level of 250 mA dc.

3. Sufficient cooling shall be supplied to maintain anode core and seal temperatures below the maximum rating, unless otherwise specified.

4. In all applications, sufficient cooling shall be supplied to make effective the requirements in note 3 above, including sufficient conduction cooling for the base contacts. In applications where reliable performance and long life are factors, operation at temperatures below the absolute maximum rating is normally beneficial.

5. In all electrical tests involving application of heater voltage, a dummy radiator or heat-sink may be attached to the anode of TUT; when this is used, forced-air cooling of the anode is permissible in order to maintain the anode core temperature below 250°C, unless otherwise specified in the test conditions. Base seals may be cooled by conduction, or by other means where required, in order to maintain seal temperatures below the maximum ratings.

6. The AQL for each test listed under quality conformance inspection, part 1, shall be 0.65, inspection level II.

7. This test shall be the first test performed after the specified holding period.

8. Cooling shall be set up to maintain the tube temperature at the hottest point on the tube envelope at 225°C ±25°C. (This setup should be permanent and need be verified only as often as required to maintain equipment accuracy.)

9. With anode, grid 1, and grid 2 tied together, apply a pulse voltage source between anode and cathode. Adjust the pulse amplitude until the specified peak cathode current is obtained. Test shall be made at the end of 1 minute or when stability is reached, whichever occurs first. The pulse voltage shall meet the following requirements:

(a)	Pulse duration	2 µs
(b)	Pulse repetition rate	60 pps
(c)	Duty factor	0.00012

10. Before subjection to this test, the tube shall be cooled to room ambient temperature for 30 minutes minimum, then preheated at Ef = 21.5 volts for 120 seconds prior to the application of other voltages. During this test, grid No. 1 shall be connected to the cathode.

11. Before subjection to this test, the tube shall be cooled to room ambient temperature for 30 minutes. Using the test circuit shown on the figure of method 1366, measure the resistance, in both directions, between any two adjacent electrodes (except across the heater terminals). If the resistance value is below that required, repeat this test after a 10-minute interval. If the tube fails again, the tube shall be rejected.

12. Tube shall be tested in a grid-driven amplifier circuit. Adjust Ec1 bias supply and tune circuit for maximum useful power output. The specified driver power (Pd) output shall be measured with a Bird Thru-line Wattmeter, or equivalent. Driving power output = forward power minus reflected power. Grid voltage supply shall have an effective impedance of 500 ohms maximum.

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NOTES: - Continued

(D) 13. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Preparing activity:
Army - ER

(Project 5960-3288)

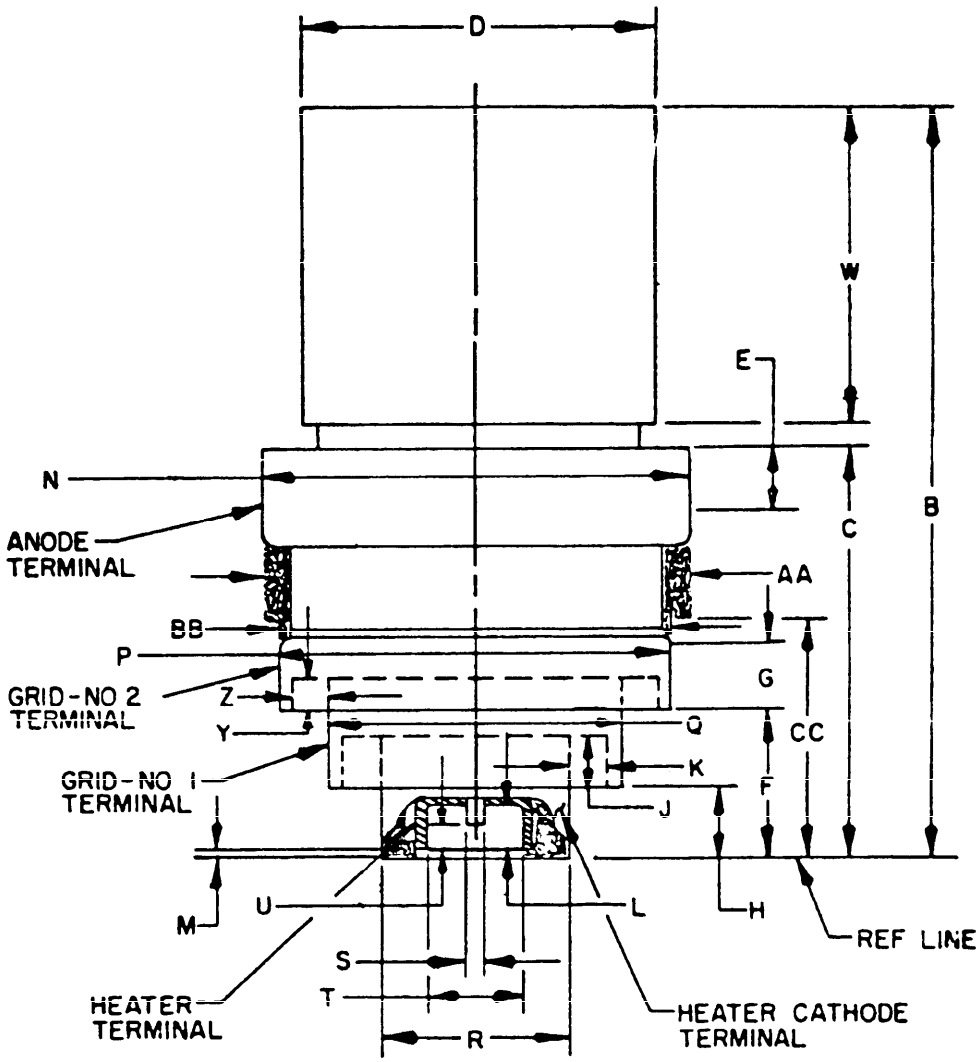
Review activities:
Air Force - 17, 99
DLA - ES

User activities:
Army - AR
Navy - AS, CG, MC, OS, SH
Air Force - 11, 19

Agent:
DLA - ES

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DEPARTMENT OF THE ARMY



Actual tube outline within these areas is not controlled by dimensions and must not be used for any purpose.

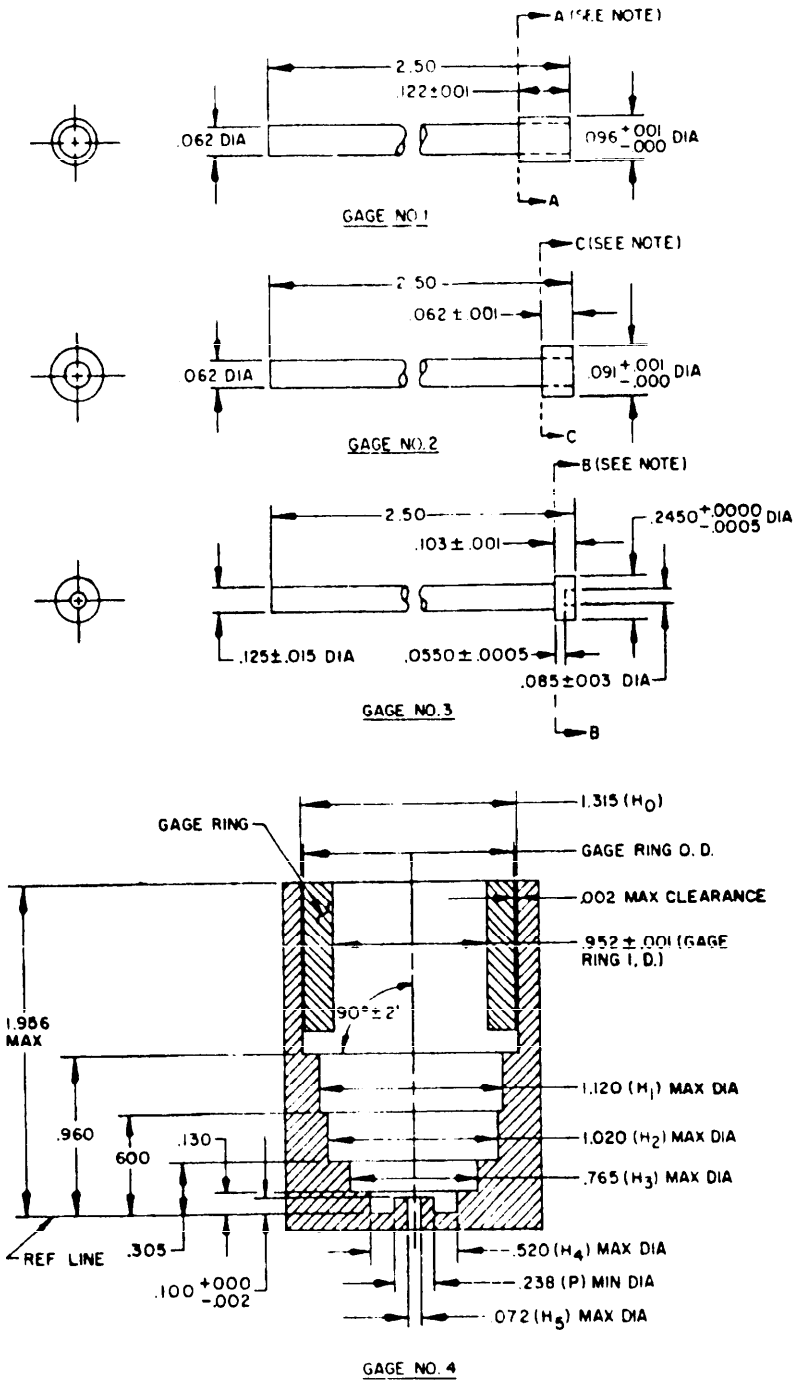
FIGURE 1. Outline drawing of electron tube type 7843.

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Dimensions in inches with metric equivalents (mm) in parentheses			
Ltr	Minimum	Maximum	Notes
	Quality conformance inspection, part 1		
	Gage check		a
	Quality conformance inspection, part 2		
B	1.805 (45.85)	1.955 (49.66)	
D	.895 (22.73)	.905 (22.99)	
	Quality conformance inspection, part 3		m
C	.990 (25.15)	1.080 (27.43)	
E	.165 (4.19)	---	
F	.340 (8.64)	.410 (10.41)	
G	.140 (3.56)	---	
H	.150 (3.81)	.200 (5.08)	
J	.120 (3.05)	---	b
K	.095 (2.41)	---	b
L	.100 (2.54)	---	d
M	---	.050 (1.27)	
N	1.085 (27.56)	---	e
P	.985 (25.02)	---	f
Q	.735 (18.67)	---	g
R	.480 (12.12)	---	h
S	---	.072 (1.83)	
T	---	.260 (6.60)	
U	.054 (1.37)	---	d
W	.780 (19.81)	---	
Y	.060 (1.52)	---	c
Z	.090 (2.29)	---	c
AA	---	---	j
BB	---	---	k
CC	.600 (15.24)	---	

FIGURE 1. Outline drawing of electron tube type 7843 - Continued.

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NOTE: The edge of the grid No. 1, No. 2, and No. 3 terminal shall be flush with or extend beyond the surface "AA", "BB", or "CC" on the gage.

INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.0005	.01	.072	1.83	.125	3.18	.765	19.43
.001	.03	.085	2.16	.130	3.30	.952	24.18
.002	.05	.091	2.31	.238	6.05	.960	24.38
.003	.08	.096	2.44	.2450	6.22	1.020	25.91
.015	.38	.100	2.54	.305	7.75	1.120	28.45
.0550	1.40	.103	2.62	.520	13.21	1.315	33.40
.062	1.57	.122	3.10	.600	15.24	1.956	49.68
						2.500	63.50

FIGURE 1. Outline drawing of electron tube type 7843 - Continued.

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NOTES:

a. With the cylindrical surfaces of the anode cap, anode terminal, screen grid terminal, control grid terminal, heater cathode terminal and heater terminal clean, smooth and free of burrs, each tube will enter the gage No. 4 to determine out of roundness, concentricity and oversize. Gage ring shall enter between gage No. 4 and anode cap of tube until insert rests on anode flange of tube. The axes of the gage ring I.D. and gage ring O.D. are coincident within .001". The axes of cylindrical holes H_0 through H_5 and the axis of post P shall be concentric within .001". Proper seating is determined by failure of a .010" thickness gage, 1/8" wide to enter between heater cathode terminal and bottom surface of H_4. A slot is provided to permit this measurement to be made. The AQL for dimensions listed under quality conformance inspection, part 1 shall be 1.0, inspection level II.

b. This dimension to be checked by gage No. 1 entering this area. The edge of the grid No. 1 terminal shall be flush with or extend beyond the surface "AA" on the gage.

c. This dimension to be checked by gage No. 2 entering this area. The edge of grid No. 2 terminal shall be flush with or extend beyond the surface "CC" on the gage.

d. This dimension to be checked by gage No. 3 entering this area. The edge of the heater terminal shall be flush with or extend beyond the surface "BB" on the gage.

e. Dimension "N" applies over length "E" only.

f. Dimension "P" applies over length "G" only.

g. Dimension "Q" applies over length "F" minus "H" only.

h. Dimension "R" applies over length "H" only.

j. On any one tube this dimension shall never be greater than "N".

k. On any one tube this dimension shall never be greater than "P".

m. Perform test(s) every six months, with sampling as follows:

n1 = 4 c1 = 0 where c2 represents the total allowable failures for the
n2 = 4 c2 = 1; first and second samples combined.

In case of failure, the failing dimension(s) shall become quality conformance inspection (QCI), part 2, AQL 6.5, inspection level S3 for three successful successive submissions, at which time the testing may revert to the twice-yearly basis.

FIGURE 1. Outline drawing of electron tube type 7843 - Continued.

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